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CAMERA MOUNTING DEVICE AND VEHICLE

Abstract

The present disclosure provides a camera mounting device that comprises: a baseplate in which at least one capture opening is disposed; and spring members, wherein each capture opening corresponds to at least two spring members, the at least two spring members are arranged to be located on two sides, respectively, of an axis of the capture opening in the baseplate. A camera is sandwiched between the at least two spring members, and a lens of the camera faces the capture opening. The camera mounting device of the disclosure does not need an extra support, thereby simplifying a mounting structure and mounting steps, reducing the cost and realizing stable securing of the camera.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to the technical field of vehicle manufacturing, and in particular, to a camera mounting device and a vehicle.

BACKGROUND

[0002] At present, in order to mount a narrow-angle camera and a wide-angle camera on a vehicle, the camera will be mounted on a bonding plate using a front camera support and a retainer, and then the bonding plate is adhered to a windshield, where a wiring harness/wire of the camera is mounted in the support. The existing camera mounting means has the following disadvantages: an extra support needs to be arranged to mount the camera to the bonding plate, leading to complicated mounting steps; the support is large and wide and occupies a large space, leading to a large size of the bonding plate; the wiring harness/wire of the camera is prolonged, needing larger space for housing; and a thermal pad inside the retainer dissipates heat, causing a bad thermal effect to the support structure.

[0003] Therefore, there is a need for an improved camera mounting device to overcome or reduce at least some of the disadvantages existing in the prior art hereinabove.

SUMMARY

[0004] In a first aspect of the disclosure, there is provided a camera mounting device which comprises: [0005] a baseplate in which at least one capture opening is disposed; and [0006] spring members, wherein each capture opening corresponds to at least two spring members, the at least two spring members are arranged to be located on two sides, respectively, of an axis of the capture opening in the baseplate; and a camera is sandwiched between the at least two spring members, and a lens of the camera faces the capture opening and aligns with the capture opening.

[0007] Optionally, the camera mounting device comprises a protruding portion that is formed by extending along a direction away from a surface of the baseplate, wherein the spring members are disposed on the protruding portion, and a recessed structure is formed on the protruding portion and is configured to house the camera.

[0008] Optionally, the protruding portion comprises a first wall and a second wall that are arranged to be located on two sides, respectively, of the axis of the capture opening in the baseplate, wherein the spring members are disposed, one on either of the first wall and the second wall, and the recessed structure is formed between the first wall and the second wall.

[0009] Optionally, a rib is formed inside the recessed structure.

[0010] Optionally, the protruding portion further comprises third walls that are formed on the sides of the first wall and the second wall, away from the capture opening, respectively, wherein the distance between a top surface of the first wall and the baseplate and the distance between a top surface of the second wall and the baseplate gradually decrease in a direction away from the third walls and towards the capture opening, so that the camera is angled with respect to the baseplate.

[0011] Optionally, the distance between the first wall and the second wall is 26.5 mm-28 mm; and/or [0012] an included angle between the camera and the baseplate is 21°-23°.

[0013] Optionally, the spring members are flat spring members.

[0014] Optionally, each of the flat spring members comprises a first section and a second section, the first section of each of the flat spring members extends along a direction away from the surface of the baseplate, and the second section of each of the flat spring members extends from an end of the first section of the flat spring member towards the flat spring member on the other side of the axis of the capture opening.

[0015] In a second aspect of the disclosure, there is provided a vehicle which comprises the aforesaid camera mounting device.

[0016] Optionally, the vehicle comprises a windshield, and the camera mounting device is attached to the windshield via the baseplate.

[0017] The camera mounting device of the disclosure is arranged to comprise the baseplate and the spring members, and the at least two spring members are arranged to be located on two sides, respectively, of the axis of the capture opening in the baseplate, so that the camera can be directly attached to the baseplate by being sandwiched between the at least two spring members. Hence, no extra support needs to be disposed, thereby simplifying the mounting structure and mounting steps and reducing the cost. In addition, the camera is sandwiched and secured between the spring members, so that the camera is stably secured. With the deformation of the spring members, the camera can be easily removed from the baseplate. Furthermore, the camera is directly attached to the baseplate, so that no thermal effect is generated, while the wiring arrangement of the camera is simpler.

[0018] Specific embodiments of the disclosure will be described in detail below with reference to the drawings. Those skilled in the art will further understand the described and other purposes, advantages and features of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Features, advantages and exemplary implementations of the disclosure will be described below with reference to the drawings. In the drawings, the same reference numerals indicate the same elements, wherein:

[0020] FIG. 1 is a schematic structural diagram of a camera mounting device according to an embodiment of the disclosure on a windshield during use.

[0021] FIG. 2 is a perspective structural diagram of the camera mounting device of FIG. 1 on the windshield during use.

[0022] FIG. 3 is a partial exploded perspective structural diagram of the camera mounting device of FIG. 1.

[0023] FIG. 4 is a schematic diagram of a process of mounting a camera to the camera mounting device of FIG. 1.

[0024] FIG. 5 is another schematic diagram of a process of mounting a camera to the camera mounting device of FIG. 1.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0025] Exemplary implementations of the disclosure will be described below in detail with reference to the drawings. The description of the exemplary implementation is merely illustrative and in no way serves as any limitation to the disclosure and application or usage thereof. Moreover, the size and scale of each component in the drawings are merely schematic and in no way strictly correspond to actual products.

[0026] The disclosure provides a camera mounting device. The camera mounting device comprises: a baseplate **101** and spring members **103**. At least one capture opening **110** is disposed in the baseplate **101** (see FIG. 3). Each capture opening **110** corresponds to at least two spring members **103**, the at least two spring members **103** are arranged to be located on two sides, respectively, of an axis of the capture opening **110** in the baseplate **101**; and a camera **102** is sandwiched between the at least two spring members **103**, and a lens of the camera **102** faces the capture opening **110** and aligns with the capture opening **110**. The camera mounting device of the embodiment of the disclosure is arranged to comprise the baseplate **101** and the spring members **103**, and the at least two spring members **103** are arranged to be located on two sides, respectively, of the axis of the capture opening **110** in the baseplate **101**, so that the camera **102** can be directly attached to the baseplate **101** by being sandwiched between the at least two spring members **103**. Hence, no extra

support needs to be disposed, thereby simplifying the mounting structure and mounting steps and reducing the cost. In addition, the camera **102** is sandwiched and secured between the spring members **103**, so that the camera **102** is stably secured. An extra protective cover or the like needs not to be added on the camera **102**. With the deformation of the spring members **103**, the camera **102** can be easily removed from the baseplate **101**. Moreover, the camera **102** is directly attached to the baseplate **101**, so that no thermal effect is generated, and the wiring arrangement of the camera **102** is simpler. It can be understood that the camera mounting device is applicable to any apparatus or transportation means in which the camera **102** needs to be mounted, including, but not limited to, a vehicle, an autonomous mobile robot, a self-service robot, etc.

[0027] The disclosure further provides a vehicle that has the aforesaid camera mounting device. In particular, the vehicle comprises a windshield **200**, and the camera mounting device is attached to the windshield **200** via the baseplate **101**. The baseplate **101** is generally adhered to the windshield **200**. It can be understood that the baseplate **101** may also be attached by other fixation modes, for example, suctioned to the windshield **200** via a suction disk. By adhering the camera mounting device to the windshield **200**, the stable and simple arrangement of the front camera **102** can be realized. The vehicle may be a vehicle with an advanced driver assistance system (ADAS), or vehicles other than a vehicle with an ADAS.

[0028] Constitution or the like of the camera mounting device of the embodiment of the disclosure will be introduced below in further detail with reference to the drawings. In each drawing, the windshield **200** is only partially shown.

[0029] For ease of description, in FIG. 2, an X direction, a Y direction and a Z direction are shown. In an example where the camera mounting device is adhered to the windshield **200** of the vehicle, the X direction is roughly parallel to a vertical height direction of the windshield **200**, the Y direction is roughly parallel to a horizontal length direction of the windshield **200**, and the Z direction is roughly perpendicular to the X direction and the Y direction. In addition, an included angle α between the axis of the capture opening **110** and the X direction is roughly shown by dotted lines in FIG. 2, which is also an included angle α between the axis of the camera **102** and the baseplate **101**. Furthermore, the descriptions such as “upper”, “lower”, “left” and “right” used herein also take the example of adhering the camera mounting device to the windshield **200** of the vehicle.

[0030] The number of capture openings **110** disposed in the baseplate **101** may be one or more, for example, one, two or four. For each capture opening **110**, one camera **102** is disposed correspondingly. The camera **102** may be generally classified into a narrow-angle camera or a wide-angle camera.

[0031] Each capture opening **110** corresponds to at least two spring members **103** that are arranged to be located on two sides, respectively, of an axis of the capture opening **110** in the baseplate **101**. In other words, in the Y direction, at least one spring member **103** should be disposed on each of left and right sides of the capture opening **110**, such that when the camera **102** is sandwiched between these spring members **103**, a lens of the camera **102** can face the capture opening **110** and align with the capture opening **110**. The spring members **103** hold the camera **102** mainly in the Z direction. In general, the lens of the camera **102** directly faces the capture opening **110**, and the axis of the capture opening **110** overlaps the axis of the camera **102**. For example, each capture opening **110** corresponds to two spring members **103**, one of which is located on the left side of the axis of the capture opening **110**, and the other one of which is located on the right side of the axis of the capture opening **110**. For another example, each capture opening **110** corresponds to three spring members **103**, two of which are located on the left side of the axis of the capture opening **110**, and the rest one of which is located on the right side of the axis of the capture opening **110**.

[0032] In the camera mounting device of the embodiment shown in FIGS. 1 and 2, two capture openings **110** are disposed in the baseplate **101**. Above each capture opening **110**, two spring members **103** are symmetrically disposed on the left and right sides, respectively, of the axis of the

capture opening **110**. Each of the two cameras **102** is sandwiched between the two corresponding spring members **103**, and the lens of each camera **102** faces its corresponding capture opening **110**. The two cameras **102** may be a narrow-angle camera and a wide-angle camera respectively, which are used for visual monitoring in front of the vehicle.

[0033] In some embodiments, the spring members **103** may be flat spring members. With the flat spring members **103**, the distance between the two flat spring members disposed oppositely can increase by bending the flat spring members outwards, so that the camera **102** is easy to remove. Meanwhile, the flat spring members have simple structures and thus can be easily fixed to the baseplate **101**. In general, the spring members **103** that are made of a metal material are aesthetics and durable. It can be understood that the spring members **103** made of other materials may also be used.

[0034] FIG. **3** is a partial exploded perspective structural diagram of the camera mounting device of FIG. **1**. Referring to FIG. **3**, in a preferred implementation, each flat spring member may comprise a first section **131** and a second section **132**, the first section **131** of each flat spring member extends along a direction away from the surface of the baseplate **101**, and the second section **132** of each flat spring member extends from an end of the first section **131** of the flat spring member towards the flat spring member on the other side of the axis of the capture opening **110**. Each flat spring member is arranged to comprise the first section **131** and the second section **132**. Once the camera **102** is inserted between the two flat spring members facing each other, the second sections **132** of the flat spring members will be released on an outer surface of the camera **102** and apply pressure to the outer surface of the camera **102**. In addition, a clamping force between the two second sections **132** facing each other is large, thereby ensuring stable securing of the camera **102**. In addition, in order to further improve stable securing, bulged portions **120** may be formed on the outer surface of the camera **102**. The bulged portions **120** are configured to match with the second sections **132**, such that once the camera **102** is inserted between the two flat spring members facing each other, the second sections **132** of the flat spring members will be released on the bulged portions **120** of the camera **102** and secured together with the camera **102**. With reference to FIG. **2**, the first section **131** is formed by extending from the baseplate **101** roughly in the Z direction, and the second section **132** is formed by extending from an end of the first section **131** of one spring member **103** towards one side of the other spring member **103**. The first section **131** may be of a straight structure or a bent structure. As shown in FIG. **3**, the first section **131** preferably is of a bent structure. Compared with a straight structure, the bent first section **131** may provide higher clamping and deformation capabilities.

[0035] In some embodiments, the camera mounting device of the embodiment of the disclosure may comprise a protruding portion **104** that is formed by extending along a direction away from the surface of the baseplate **101**, wherein the spring members **103** are disposed on the protruding portion **104**, and a recessed structure **140** is formed on the protruding portion **104** and is configured to house the camera **102**. With the protruding portion **104** on which the recessed structure **140** configured to house the camera **102** is formed, the camera **102** may be properly angled with respect to the baseplate **101**. Meanwhile, with the matching between the recessed structure **140** and the spring members **103**, the camera **102** can be stabilized better. In the embodiment shown in FIGS. **1** and **2**, the camera mounting device comprises two protruding portions **104**, with one protruding portion **104** corresponding to one capture opening **110** and one camera **102**. The protruding portion **104** is formed by extending towards one side of the interior of the vehicle roughly in the Z direction.

[0036] As can be seen from FIGS. **2** and **3**, the protruding portion **104** may comprise a first wall **141** and a second wall **142** that are arranged to be located on two sides, respectively, of the axis of the capture opening **110** in the baseplate **101**, wherein the spring members **103** are disposed, one on either of the first wall **141** and the second wall **142**, and the recessed structure **140** is formed between the first wall **141** and the second wall **142**. With the first wall **141** and the second wall **142**

that match with the spring members **103**, the position of the camera **102** on the baseplate **101** can remain unchanged, thereby firmly securing the camera **102** and not allowing the camera **102** to move. It can be understood that adjacent protruding portions **104** may share a wall when a plurality of protruding portions **104** are disposed. The spring members **103** and the protruding portion **104** are assembled into a whole. Referring to FIG. 3, clamp slots may be provided in the first wall **141** and the second wall **142**. A clamp rod is disposed in each clamp slot. The flat spring members are bent through the clamp rods in the clamp slots, so that the flat spring members are fixed to the protruding portion **104**.

[0037] In some embodiments, a rib **144** may be formed inside the recessed structure **140**. The rib **144** helps to slide and securing of the surface of the camera **102**. In order to better fit with the camera **102**, an arc-shaped surface is generally formed inside the recessed structure **140**. The recessed structure **140** may form the arc-shaped surface as a whole, or a structure with an arc-shaped surface may be formed inside the recessed structure **140**. In the embodiment shown in FIG. 3, a structure with an arc-shaped surface is formed inside the recessed structure **140** by extending between the first wall **141** and the second wall **142** along a direction away from the surface of the baseplate **101**, and a rib **144** is formed on the structure. At the recessed structure **140**, a plurality of ribs **144**, for example, three ribs, may be disposed at intervals between the first wall **141** and the second wall **142**. The rib **144** generally has a smooth top surface.

[0038] The protruding portion **104** may further comprise third walls **143** that are formed on sides of the first wall **141** and the second wall **142**, away from the capture opening **110**, respectively, wherein the distance between a top surface of the first wall **141** and the baseplate and the distance between a top surface of the second wall **142** and the baseplate **101** gradually decrease in a direction away from the third walls **143** and towards the capture opening **110**, so that the camera **102** is angled with respect to the baseplate **101**. With the third walls **143**, better stability and securing position may be provided for the camera **102**. For example, as can be seen from FIG. 3, the third walls **143** may lean against a back surface of the camera **102**. Meanwhile, the first wall **141** and the second wall **142** are configured into a slope structure, so that the camera **102** may be disposed obliquely relative to the baseplate **101**. In this way, the lens of the camera **102** is better oriented towards the capture opening **110** and aligned with the capture opening **110**. As shown in FIGS. 1 and 2, the first wall **141**, the second wall **142** and the third walls **143** are formed by extending towards one side of the interior of the vehicle roughly in the Z direction, the first wall **141** and the second wall **142** are symmetrically located on the left and right sides of the axis of the capture opening **110** along the Y direction, and the third walls **143** are formed towards each other on the sides of the first wall **141** and the second wall **142** away from the capture opening **110**.

[0039] The distance between the first wall **141** and the second wall **142** may be 26.5 mm-28 mm, for example, 26.5 mm, 27.8 mm and 28 mm. The structures of a plurality of protruding portions **104** on the same baseplate **101** may be identical. That is, the distance between the first wall **141** and the second wall **142** is the same. In this case, a mounting space for the camera is adjustable using the elasticity of the spring members **103** so as to fit with the camera **102** of different sizes. Of course, the structures of a plurality of protruding portions **104** on the same baseplate **101** may be different.

[0040] Generally, an included angle between the camera **102** and the baseplate **101** may be 21°-23°, for example, 21°, 22° and 23°.

[0041] FIG. 4 is a schematic diagram of a process of mounting a camera **102** to the camera mounting device of FIG. 1. FIG. 5 is another schematic diagram of a process of mounting a camera **102** to the camera mounting device of FIG. 1. FIGS. 1 and 2 are schematic diagrams after the camera **102** is mounted. When the camera mounting device of the embodiment of the disclosure is used in the vehicle, the camera **102** is inserted in the Z direction between the two spring members **103** and further inserted until the camera **102** is housed in the recessed structure **140**. Then, the baseplate **101** is attached to the windshield **200**, thus completing the mounting of the front camera

102. When the camera mounting device of the embodiment of the disclosure is used in the vehicle, it is also possible to first attach the baseplate **101** to the windshield **200**, then insert the camera **102** in the Z direction between the two spring members **103** and further insert until the camera **102** is housed in the recessed structure **140**, thus completing the mounting of the front camera **102**. In FIG. **4**, the camera **102** is to be inserted between the two spring members **103**. In FIG. **5**, the camera **102** is already inserted between the two spring members **103** and housed in the recessed structure **140**.

[0042] In the description of this specification, the reference to the terms “one implementation”, “some implementations”, “one embodiment”, “some embodiments” or “preferred implementations” means that the specific features, structures, materials or characteristics described in view of the embodiment or example are included in at least one embodiment or example of the disclosure. In this specification, the schematic expressions of these terms do not necessarily refer to the same embodiment or example. Moreover, the specific features, structures, materials, or characteristics described can be combined in any one or more embodiments or examples in an appropriate manner.

[0043] Implementations of the disclosure are described in detail above. However, the aspects of the disclosure are not limited to the implementations hereinabove. Various modifications and substitutions may be applied to the implementations hereinabove without departing from the scope of the disclosure.

Claims

1. A camera mounting device, wherein the camera mounting device comprises: a baseplate in which at least one capture opening is disposed; and spring members, wherein each capture opening corresponds to at least two spring members, the at least two spring members are arranged to be located on two sides, respectively, of an axis of the capture opening in the baseplate; and a camera is sandwiched between the at least two spring members, and a lens of the camera faces the capture opening and aligns with the capture opening.
2. The camera mounting device according to claim 1, wherein: the camera mounting device comprises a protruding portion that is formed by extending along a direction away from a surface of the baseplate, wherein the spring members are disposed on the protruding portion, and a recessed structure is formed on the protruding portion and is configured to house the camera.
3. The camera mounting device according to claim 2, wherein: the protruding portion comprises a first wall and a second wall which are arranged to be located on two sides, respectively, of the axis of the capture opening in the baseplate, wherein the spring members are disposed, one on either of the first wall and the second wall, and the recessed structure is formed between the first wall and the second wall.
4. The camera mounting device according to claim 2, wherein: a rib is formed inside the recessed structure.
5. The camera mounting device according to claim 3, wherein: the protruding portion further comprises third walls which are formed on the sides of the first wall and the second wall, away from the capture opening, respectively, wherein the distance between a top surface of the first wall and the baseplate and the distance between a top surface of the second wall and the baseplate gradually decrease in a direction away from the third walls and towards the capture opening, so that the camera is angled with respect to the baseplate.
6. The camera mounting device according to claim 5, wherein: the distance between the first wall and the second wall is 26.5 mm-28 mm; and/or an included angle between the camera and the baseplate is 21°-23°.
7. The camera mounting device according to claim 1, wherein the spring members are flat spring members.
8. The camera mounting device according to claim 7, wherein: each of the flat spring members

- comprises a first section and a second section, the first section of each of the flat spring members extends along a direction away from the surface of the baseplate, and the second section of each of the flat spring members extends from an end of the first section of the flat spring member towards the flat spring member on the other side of the axis of the capture opening.
- 9.** A vehicle, wherein the vehicle has a camera mounting device, and the camera mounting device comprises: a baseplate in which at least one capture opening is disposed; and spring members, wherein each capture opening corresponds to at least two spring members, the at least two spring members are arranged to be located on two sides, respectively, of an axis of the capture opening in the baseplate; and a camera is sandwiched between the at least two spring members, and a lens of the camera faces the capture opening and aligns with the capture opening.
- 10.** The vehicle according to claim 9, wherein: the vehicle comprises a windshield, and the camera mounting device is attached to the windshield via the baseplate.
- 11.** The vehicle according to claim 9, wherein: the camera mounting device comprises a protruding portion that is formed by extending along a direction away from a surface of the baseplate, wherein the spring members are disposed on the protruding portion, and a recessed structure is formed on the protruding portion and is configured to house the camera.
- 12.** The vehicle according to claim 11, wherein: the protruding portion comprises a first wall and a second wall which are arranged to be located on two sides, respectively, of the axis of the capture opening in the baseplate, wherein the spring members are disposed, one on either of the first wall and the second wall, and the recessed structure is formed between the first wall and the second wall.
- 13.** The vehicle according to claim 11, wherein: a rib is formed inside the recessed structure.
- 14.** The vehicle according to claim 12, wherein: the protruding portion further comprises third walls which are formed on the sides of the first wall and the second wall, away from the capture opening, respectively, wherein the distance between a top surface of the first wall and the baseplate and the distance between a top surface of the second wall and the baseplate gradually decrease in a direction away from the third walls and towards the capture opening, so that the camera is angled with respect to the baseplate.
- 15.** The vehicle according to claim 14, wherein: the distance between the first wall and the second wall is 26.5 mm-28 mm; and/or an included angle between the camera and the baseplate is 21°-23°.
- 16.** The vehicle according to claim 9, wherein the spring members are flat spring members.
- 17.** The vehicle according to claim 16, wherein: each of the flat spring members comprises a first section and a second section, the first section of each of the flat spring members extends along a direction away from the surface of the baseplate, and the second section of each of the flat spring members extends from an end of the first section of the flat spring member towards the flat spring member on the other side of the axis of the capture opening.
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